Uncertainty Calibration of Multi-Label Bird Sound Classifiers

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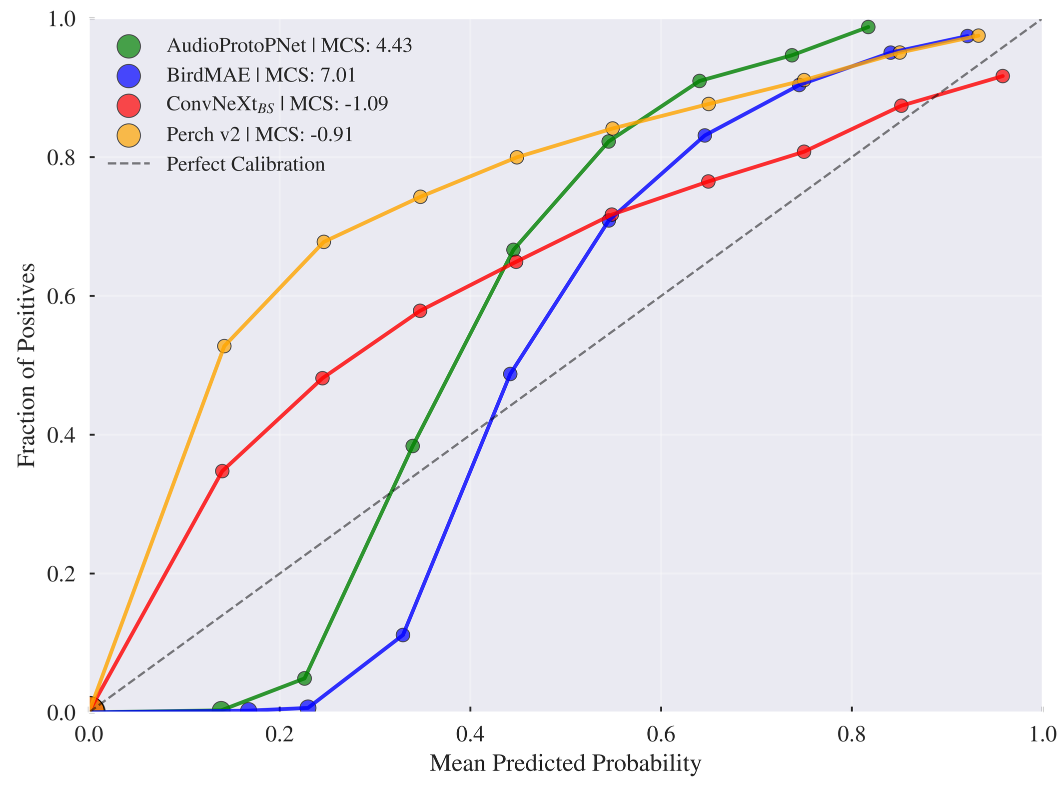
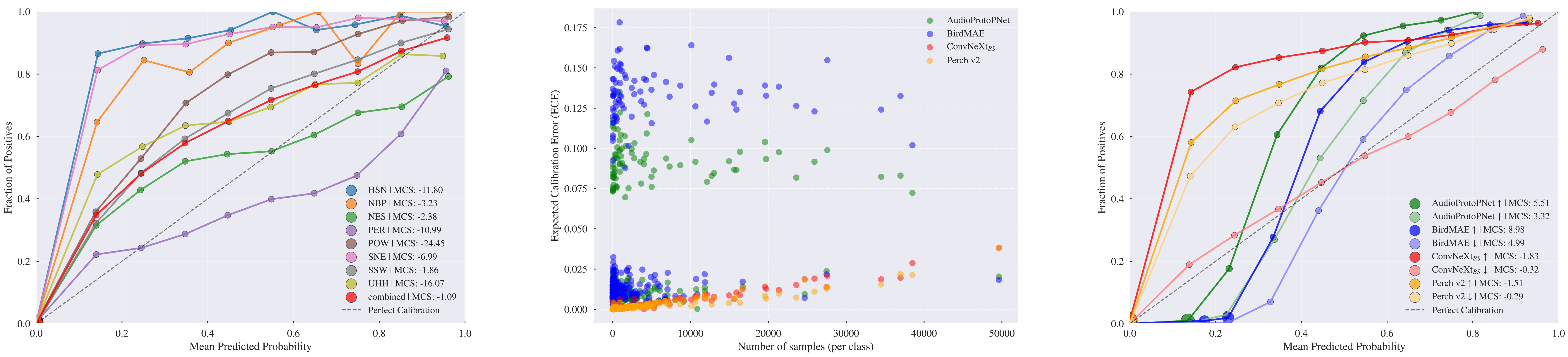


Figure 1: Reliability diagrams of multi-label bird sound classifiers on BirdSet [1].

Contributions

- **C1. Benchmark calibration** of four state-of-the-art multi-label bird sound classifiers on BirdSet [1], reporting global, per-dataset and per-class calibration with threshold-free metrics
- **C2. Investigate post hoc strategies** (temperature [4] and Platt scaling [5]): Platt scaling with 10 minutes of test data can significantly improve calibration
- **C3. Summarise implications** to give practition- ers concrete suggestions for deployment specific calibration evaluation and improvement.

Benchmarking the Calibration of Bird Sound Classifiers



(a) Per-dataset reliability diagram for ConvNeXt$_{\mathrm{BS}}$.

(b) Class-wise ECE vs. sample count.

(c) Reliability of most vs. least frequent classes.

Experimental setting:

- **Models:** AudioProtoPNet, BirdMAE [2], ConvNeXt$_{\mathrm{BS}}$ [1], Perch v2 [3] (multi-label classifiers on BirdSet [1]).
- **BirdSet:** Eight expert-annotated test datasets from PAM deployments.
- **Metrics:** ECE (expected calibration error), MCS (miscalibration score), OCS/UCS (over/underconfidence); cmAP for discrimination.

Results:

- Globally, Perch v2 and ConvNeXt$_{\mathrm{BS}}$ are underconfident; AudioProtoPNet and BirdMAE are overconfident (Fig. 2a).
- *Calibration varies significantly per dataset* (Fig. 2a); combined ECE can mask domain-specific miscalibration.
- Class-wise ECE (Fig. 2b) shows Perch v2 and ConvNeXt$_{\mathrm{BS}}$ with low ECE, increasing with numbers of samples per class.
- For common vs. rare species (Fig. 2c), rare classes shift toward overconfidence for AudioProtoPNet and BirdMAE; Perch v2 and ConvNeXt$_{\mathrm{BS}}$ remain underconfident for both subsets.

Post-hoc Calibration

- **Temperature scaling** (TS) and **Platt scaling** (PS) evaluated with two calibration sets: global parameters on POW and per-class parameters on the first 10 minutes of each test dataset.
- Per-class Platt scaling with 10 minutes of labelled data provides the most consistent calibration improvement across models and datasets.

Table 1: Median relative MCS improvement (%) across test sets.

Model	TS POW (%)	PS POW (%)	TS/cl. (%)	PS/cl. (%)
AudioProtoPNet	56	56	75	**92**
BirdMAE	81	89	94	**95**
ConvNeXt$_{\mathrm{BS}}$	10	20	**67**	12
Perch v2	33	24	23	**61**

Conclusions

- Calibration varies substantially across datasets and classes; aggregating across domains can mask systematic miscalibration
- Perch v2 [3] and ConvNeXt$_{\mathrm{BS}}$ [1, **?**] consistently underconfident; AudioProtoPNet and BirdMAE [2] show mixed over-/underconfidence
- Simple post hoc methods (Platt scaling [5] with 10 min labelled data) can significantly improve deployment-specific calibration
- Recommend reporting OCS/UCS alongside ECE or MCS to reveal whether errors stem from over- or underconfidence
- Future work: Bayesian approaches, calibration@k metrics, and evaluation across additional deployment scenarios

KEY REFERENCES

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